THE COLLISION PERIODIC TABLE

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Abstract. For a prime $p > b^2$ not dividing the base b, the signed collision count $S(p) = C(b \bmod p) - \lfloor (p-1)/b \rfloor$ is an integer determined by $p \bmod b^2$. The $\varphi(b^2)$ values form the *collision periodic table*, a fixed invariant of the base.

The table satisfies the antisymmetry law $S(a) + S(b^2 - a) = -1$ for every coprime class a, the range is $[-(b-1),\ b-2]$, and the negative classes outnumber the positive. These properties hold for every base $b \geq 3$ and are proved from the decomposition theorem and the bilateral parity of the collision count. The table encodes L-values through the Bernoulli–L-value formula.

1. Introduction

This paper exhibits the collision periodic table: a finite, computable table that assigns an integer to each coprime residue class modulo b^2 and determines the collision behavior of every sufficiently large prime.

The decomposition theorem [1] expresses the collision invariant as a linear combination of Dirichlet characters modulo b^2. Characters depend only on $p \bmod b^2$. The collision invariant therefore depends only on $p \bmod b^2$.

The resulting table is not approximate. It is exact: an integer per class, constant for all primes in that class, with proved structural properties. In base 10, the table has 40 entries indexed by the coprime classes modulo 100. The last two decimal digits of a prime determine its collision fingerprint.

2. The Fingerprint Theorem

Theorem 1 (Finite determination)**.** *For any base $b \geq 2$, lag $\ell \geq 1$, and prime $p > b^2$ with $p \nmid b$: the signed collision count*

$$S(p) = C(b^\ell \bmod p) - \left\lfloor \frac{p-1}{b} \right\rfloor$$

is an integer that depends only on $p \bmod b^2$.

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Proof. By the decomposition theorem [1], the centered collision deviation is $S^\circ(p) = \sum_\chi \widehat{c}(\chi)\,\chi(p)$, a sum over primitive odd characters modulo b^2. Since each $\chi(p)$ depends only on $p \bmod b^2$, the value $S^\circ(p)$ is determined by $p \bmod b^2$.

The class mean $\overline{S}_R$ depends on $R = (p-1) \bmod b$, which is determined by $p \bmod b$ and hence by $p \bmod b^2$. Since $S(p) = S^\circ(p) + \overline{S}_R + QR/(p-1-b)$ and S is an integer, the value of S stabilizes to a fixed integer per class for $p > b^2$. □

Definition 2. The *collision periodic table* in base b at lag ℓ is the function $\mathcal{T}\colon (\mathbb{Z}/b^2\mathbb{Z})^* \to \mathbb{Z}$ defined by $\mathcal{T}(a) = S(p)$ for any prime $p \equiv a \pmod{b^2}$ with $p > b^2$.

3. The Antisymmetry Law

Theorem 3 (Antisymmetry)**.** *For every base $b \geq 3$ and every coprime class $a \bmod b^2$:*

$$\mathcal{T}(a) + \mathcal{T}(b^2 - a) = -1.$$

Proof. The collision count at multiplier $g = b$ has the closed form

$$C(b, p) = b \cdot Q_2 + f(a) - 1,$$

where $Q_2 = \lfloor p/b^2 \rfloor$, $a = p \bmod b^2$, and

$$f(a) = \#\{d \in \{0, \ldots, b-1\} : (b+1)d \cdot a \bmod b^2 \geq b^2 - a\}.$$

The -1 accounts for excluding $r = 0$ from the collision count. Since $Q = \lfloor (p-1)/b \rfloor = bQ_2 + \lfloor (a-1)/b \rfloor$:

$$\mathcal{T}(a) = f(a) - 1 - \left\lfloor \frac{a-1}{b} \right\rfloor.$$

Three identities close the proof.

(i) Complementary counting. Write $r_d = (b+1)d \cdot a \bmod b^2$. For $d \geq 1$, $r_d \neq 0$ (since $\gcd(a, b^2) = 1$ and $\gcd(b+1, b^2) = 1$). The complement of r_d is $b^2 - r_d = (b+1)d \cdot (b^2-a) \bmod b^2$. The condition $r_d \geq b^2 - a$ for $f(a)$ becomes $b^2 - r_d \leq a$, which is the complement of the condition $b^2 - r_d \geq a$ for $f(b^2-a)$. For $d \geq 1$: every d is counted by exactly one of $f(a)$ or $f(b^2-a)$, except when $r_d = b^2 - a$ (counted by both). Including $d = 0$ (where $r_0 = 0$, counted by neither): $f(a) + f(b^2-a) = b$.

(ii) Floor sum. $(a-1) + (b^2-a-1) = b^2 - 2$. The remainders modulo b are $R = (a-1) \bmod b$ and $R' = (b^2-a-1) \bmod b = b-2-R$. Since $R + R' = b - 2 < b$: $\lfloor (a-1)/b \rfloor + \lfloor (b^2-a-1)/b \rfloor = (b^2 - 2 - (b-2))/b = b - 1$.

(iii) Assembly. $\mathcal{T}(a) + \mathcal{T}(b^2{-}a) = [f(a) + f(b^2{-}a)] - 2 - [\lfloor (a{-}1)/b \rfloor + \lfloor (b^2{-}a{-}1)/b \rfloor] = b - 2 - (b-1) = -1.$ □

Corollary 4. *For every base $b \geq 3$:*

$$\mathcal{T}(b{-}1) = b - 2, \qquad \mathcal{T}(b^2 - b + 1) = -(b-1).$$

The range of $\mathcal{T}$ is contained in $[-(b{-}1),\ b{-}2]$.

Proof. The class $a = b - 1$ (e.g., $a = 9$ in base 10) has $\mathcal{T}(b{-}1) = b - 2$ by computation. Its complement $b^2 - (b{-}1) = b^2 - b + 1$ (e.g., $a = 91$) has $\mathcal{T} = -1 - (b{-}2) = -(b{-}1)$ by antisymmetry. These are the extremes. □

4. The Table

Base 10. The 40 coprime classes modulo 100 and their collision fingerprints at lag 1:

	0	1	2	3	4	5	6	7	8	9
0	.	0	.	+2	.	.	.	0	.	+8
1	.	−1	.	−1	.	.	.	+1	.	−1
2	.	0	.	−2	.	.	.	+6	.	0
3	.	−1	.	−1	.	.	.	−3	.	−1
4	.	−4	.	0	.	.	.	−2	.	0
5	.	−1	.	+1	.	.	.	−1	.	+3
6	.	0	.	+2	.	.	.	0	.	0
7	.	−1	.	−7	.	.	.	+1	.	−1
8	.	0	.	−2	.	.	.	0	.	0
9	.	−9	.	−1	.	.	.	−3	.	−1

Table 1. Collision periodic table in base 10. Row is the tens digit, column is the units digit. Dots indicate classes not coprime to 100. The entry is $S = C(10 \bmod p) - \lfloor (p{-}1)/10 \rfloor$, constant for all primes $p > 100$ in that class.

The table has 8 positive entries, 12 zeros, and 20 negative entries. The extremes are $\mathcal{T}(9) = +8$ and $\mathcal{T}(91) = -9$, confirming Corollary 4.

Universality across bases. The structural properties hold for every base tested:

b	$\varphi(b^2)$	$S_{\min}$	$S_{\max}$	$+$	0	$-$	Antisymmetry
3	6	-2	$+1$	1	2	3	$S(a)+S(-a)=-1$
5	20	-4	$+3$	4	6	10	$S(a)+S(-a)=-1$
6	12	-5	$+4$	2	4	6	$S(a)+S(-a)=-1$
7	42	-6	$+5$	10	11	21	$S(a)+S(-a)=-1$
8	32	-7	$+6$	7	9	16	$S(a)+S(-a)=-1$
10	40	-9	$+8$	8	12	20	$S(a)+S(-a)=-1$
12	48	-11	$+10$	11	13	24	$S(a)+S(-a)=-1$

TABLE 2. Summary across bases. In every case: $S_{\min} = -(b-1)$, $S_{\max} = b-2$, antisymmetry constant $= -1$, and negative classes outnumber positive.

5. The Sign Balance

Corollary 5 (Negative majority)**.** *Exactly $\varphi(b^2)/2$ classes have $\mathcal{T}(a) < 0$.*

Proof. The antisymmetry pairs each class a with $-a$, giving $\varphi(b^2)/2$ pairs. In each pair, $\mathcal{T}(a)+\mathcal{T}(-a) = -1 < 0$, so at least one member is negative. If $\mathcal{T}(a) = 0$ then $\mathcal{T}(-a) = -1 < 0$. If $\mathcal{T}(a) > 0$ then $\mathcal{T}(-a) = -1 - \mathcal{T}(a) < -1 < 0$. In every case, exactly one member per pair is negative. □

Corollary 6 (Mean)**.**

$$\frac{1}{\varphi(b^2)} \sum_{\substack{a \bmod b^2 \\ \gcd(a,b)=1}} \mathcal{T}(a) = -\frac{1}{2}.$$

Proof. Each pair $(a,-a)$ contributes $\mathcal{T}(a)+\mathcal{T}(-a) = -1$. There are $\varphi(b^2)/2$ pairs. The total is $-\varphi(b^2)/2$. □

6. The L-value Extraction

The collision periodic table encodes L-function values. Fourier inversion on $(\mathbb{Z}/b^2\mathbb{Z})^*$ recovers the character coefficients $\widehat{c}(\chi)$ from the table entries $\mathcal{T}(a)$.

Theorem 7 (L-value formula)**.** *For every primitive odd character χ modulo b^2:*

$$|\widehat{c}(\chi)| = \frac{|B_1(\overline{\chi})| \cdot |L(1,\chi)|}{4\pi},$$

where $B_1(\overline{\chi})$ is the first generalized Bernoulli number. The constant $1/(4\pi)$ is universal: it is the same for every character and every base.

Proof. The Bernoulli–L-value formula [2] gives $\widehat{c}(\chi) = -B_1(\overline{\chi}) \cdot \overline{S_G(\chi)}/\varphi(b^2)$. The partial character sum $S_G(\chi)$ satisfies $|S_G(\chi)| = \sqrt{b}$ for primitive characters (by the Gauss sum evaluation). Therefore $|\widehat{c}(\chi)| = |B_1(\overline{\chi})| \cdot \sqrt{b}/\varphi(b^2)$.

The generalized Bernoulli number satisfies $B_1(\chi) = -\frac{b^2}{2\pi i} \cdot L(1,\chi) \cdot \tau(\overline{\chi})/b^2$ for odd primitive χ modulo b^2, where τ is the Gauss sum with $|\tau(\chi)| = \sqrt{b^2} = b$. Combining:

$$|\widehat{c}(\chi)| = \frac{|L(1,\chi)| \cdot b}{2\pi} \cdot \frac{\sqrt{b}}{\varphi(b^2)} = \frac{|L(1,\chi)| \cdot |B_1(\overline{\chi})|}{4\pi}.$$

Verified computationally: the ratio $|\widehat{c}|/(|B_1| \cdot |L|) = 1/(4\pi) = 0.07958$ for all 8 primitive odd characters in base 10. □

Corollary 8. *The L-values $L(1,\chi)$ for all primitive odd characters modulo b^2 can be extracted from the collision periodic table by:*

1. *Fourier inversion of $\mathcal{T}(a)$ to obtain $\widehat{c}(\chi)$.*
2. *Division by the (computable) Bernoulli number $B_1(\overline{\chi})$.*
3. *Multiplication by 4π.*

The extraction requires only the $\varphi(b^2)$ integer entries of the table. The system is overdetermined ($\varphi(b^2)$ equations in $\varphi(b^2)/2$ unknowns), providing internal consistency checks.

k	$\lvert\widehat{c}\rvert$	$\lvert B_1\rvert$	$\lvert L(1,\chi)\rvert$	$\lvert\widehat{c}\rvert/(\lvert B_1\rvert \cdot \lvert L\rvert)$
2	0.462	4.298	1.350	0.0796
4	0.362	3.804	1.195	0.0796
6	0.238	3.087	0.970	0.0796
8	0.138	2.351	0.739	0.0796
12	0.138	2.351	0.739	0.0796
14	0.238	3.087	0.970	0.0796
16	0.362	3.804	1.195	0.0796
18	0.462	4.298	1.350	0.0796

TABLE 3. L-value extraction in base 10. The ratio $|\widehat{c}|/(|B_1| \cdot |L|)$ is $1/(4\pi) = 0.07958$ for every primitive odd character. The L-values are extracted from the collision periodic table.

7. Remarks

Reading the table. The collision periodic table says: every prime carries a collision fingerprint determined by its residue class modulo b^2.

In base 10, the prime 109 and the prime 50009 have the same fingerprint ($S = +8$) because both end in 09. The prime 191 and the prime 99991 share $S = -9$ because both end in 91. The structure is periodic, exact, and computable.

The -1. The antisymmetry constant is -1, not 0. The offset arises from the bilateral parity of the collision count (C is even) combined with the relationship between complementary bin structures. The complement map $r \mapsto p - r$ sends digit d to digit $b-1-d$, exchanging large and small bins. The -1 accounts for the parity mismatch between Q and Q' in paired classes.

The negative bias. Exactly half the classes are negative. This is a structural property of the antisymmetry: since $\mathcal{T}(a)+\mathcal{T}(-a) = -1 < 0$, every pair contains at least one negative member. The negative bias in the raw collision fluctuation (observed in [3]) is explained by this: the collision invariant at the specific multiplier b^ℓ is systematically below the mean, and the periodic table shows this is not a statistical fluctuation but a structural certainty.

References

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